Characterizing the Reliability Boundary Between Retrieval Relevance and Evidence Sufficiency in Local-First Personal Memory RAG

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Abstract—Personal knowledge management systems based on retrieval-augmented generation (RAG) must not only retrieve relevant information but also determine whether the retrieved evidence is sufficient to support an answer. This study evaluates retrieval reliability in PAM, a local-first personal-memory RAG system, with particular emphasis on this distinction. PAM combines dense vector retrieval, BM25 lexical retrieval, and Reciprocal Rank Fusion for heterogeneous personal knowledge sources. A frozen evaluation corpus containing 24 sources, 195 indexed chunks, and 199 queries was used to evaluate retrieval effectiveness, abstention, reranking, and evidence-verification mechanisms under predefined effectiveness, false-negative, false-positive, and latency constraints. The frozen hybrid baseline achieved Hit@5 of 0.9236, MRR of 0.8773, and p95 retrieval latency of 47.1 ms, with zero false negatives under the evaluated configuration. However, its false-positive rate on negative queries was 0.8571. Failure analysis showed that 31 of 36 false positives fell within the study's content-sufficiency failure analysis, including cases in which retrieved information was topically relevant but insufficient to answer the query. Cross-encoder reranking and LLM-based evidence verification were subsequently evaluated. Reranking did not provide a sufficient overall improvement, while verification reduced false positives at the cost of unacceptable false-negative rates and/or latency under the study constraints. The results indicate that retrieval relevance and answer sufficiency constitute distinct reliability problems in local-first personal RAG. The study therefore identifies evidence sufficiency as a key boundary for future abstention mechanisms rather than treating retrieval confidence alone as evidence that an answer is supported.

Index Terms—retrieval-augmented generation, personal knowledge management, hybrid retrieval, BM25, dense retrieval, reranking, abstention, evidence sufficiency, local-first AI.

I. INTRODUCTION

Personal information is increasingly distributed across notes, documents, source code, research material, transcripts, and other heterogeneous files. Retrieval-augmented generation (RAG) provides a practical way to combine a knowledge store with language generation by retrieving relevant evidence at query time rather than relying only on information encoded in model parameters [1]. For personal knowledge systems, however, retrieval quality alone is not sufficient: a retrieved passage may be highly relevant to the topic of a question while still failing to contain the specific fact required to answer it.

This distinction creates a reliability problem for personal RAG. A system that always returns its highest-scoring passages may perform well on in-scope queries while still accepting queries for which the requested information is absent from the knowledge base. Dense retrieval captures semantic similarity [2], while lexical methods such as BM25 provide sensitivity to exact or overlapping terms [3]. Hybrid retrieval can combine these complementary signals, and Reciprocal Rank Fusion (RRF) provides a score-independent method for combining ranked lists [4]. However, these mechanisms primarily estimate retrieval relevance; they do not directly determine whether the retrieved evidence is sufficient to support a particular answer.

PAM (Personal AI Memory) is used in this study as a controlled local-first RAG system for investigating this boundary. Its evaluated retrieval pipeline combines 768-dimensional embeddings from nomic-embed-text with BM25 retrieval and RRF, followed by an abstention gate before grounded question answering with a local Ollama model. Optional cross-encoder reranking and evidence-verification mechanisms were evaluated as candidate interventions for improving reliability.

The study addresses four research questions. RQ1 asks how effectively hybrid dense and lexical retrieval retrieves relevant information from a heterogeneous personal knowledge base. RQ2 asks whether retrieval-based abstention can distinguish clearly irrelevant queries from queries whose requested information is absent from the knowledge base. RQ3 asks whether reranking and evidence-verification mechanisms can reduce false-positive acceptance while satisfying predefined effectiveness, false-negative, and latency constraints. RQ4 asks what the observed failure patterns reveal about the boundary between retrieval relevance and answer sufficiency.

The principal contribution is empirical rather than a claim of a novel retrieval algorithm. The study provides (i) a local-first hybrid personal-memory RAG architecture, (ii) a frozen evaluation of dense-plus-lexical retrieval and cosine-based abstention, (iii) controlled evaluation of reranking and answerability mechanisms under explicit guardrails, and (iv) a failure analysis showing that the dominant false-positive mode is content insufficiency rather than simple topic irrelevance. The findings are presented as evidence from a controlled personal-memory corpus and are not intended to establish universal performance across RAG systems or domains.

II. RELATED WORK

The cited literature establishes the principal components used in PAM—retrieval-augmented generation, dense retrieval, BM25, rank fusion, neural reranking, and evidence sufficiency—but these components do not by themselves resolve the boundary between relevance and answerability in a personal-memory setting. This study contributes an empirical evaluation of that boundary. It first measures a frozen hybrid retrieval baseline, then evaluates abstention, reranking, and evidence-verification interventions under explicit guardrails. The resulting analysis is intended to identify which reliability failures can be addressed by retrieval and which require evidence-sufficiency reasoning at the answering layer. Related work also includes sentence-level dense representations [7], late-interaction neural retrieval [8], hypothetical-document retrieval [9], self-reflective retrieval and generation [10], reference-free RAG evaluation [11], and studies of long-context evidence use [12]. These approaches provide relevant alternatives and evaluation perspectives, but they are not substituted for the frozen PAM baseline in this study.

A. Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) combines a parametric language model with an external non-parametric memory so that responses can be conditioned on retrieved evidence [1]. This architecture is attractive for personal knowledge systems because the external memory can be updated without retraining the language model and retrieved passages can provide provenance for generated answers. The present study adopts this architecture but focuses on a reliability question that occurs after retrieval: whether the retrieved evidence is actually sufficient to answer the user's question.

B. Dense Retrieval

Dense retrieval represents queries and passages in a learned vector space and ranks candidates according to embedding similarity [2]. Its principal advantage is semantic matching when relevant passages do not share exact vocabulary with the query. PAM uses local embedding inference with nomic-embed-text and 768-dimensional representations. Dense retrieval therefore supplies the semantic component of the evaluated hybrid search pipeline, but its similarity score is treated as a retrieval signal rather than as direct evidence of answerability.

C. Lexical Retrieval and BM25

BM25 is a probabilistic lexical retrieval function that incorporates term frequency, inverse document frequency, and document-length normalization [3]. Lexical retrieval remains useful when exact terms, identifiers, names, or domain-specific wording carry important information. PAM combines BM25 with dense retrieval using k1=1.5 and b=0.75. This complementary design is important for personal corpora, where some questions are semantic while others depend on precise terminology.

D. Hybrid Retrieval and Rank Fusion

Hybrid retrieval combines candidate rankings from different retrieval mechanisms. Reciprocal Rank Fusion (RRF) provides a simple rank-based method for combining such lists without requiring the component scores to share a common numerical scale [4]. PAM applies RRF with k=60 to merge dense and BM25 rankings. In this study, hybrid retrieval is evaluated as the primary retrieval mechanism, while later interventions test whether additional ranking or verification stages can address its false-positive acceptance behavior.

E. Neural Reranking

Neural rerankers provide a second-stage relevance model by evaluating a query and candidate passage jointly. BERT-based passage reranking demonstrated the value of such query-passage interaction for information retrieval [5]. PAM therefore evaluates cross-encoder reranking as an optional intervention after hybrid retrieval. Importantly, the study does not assume that a stronger relevance model necessarily improves system reliability; reranking is judged against predefined retrieval, false-negative, false-positive, and latency constraints.

F. Evidence Sufficiency and Abstention

A separate line of work considers whether a RAG system should answer when the retrieved evidence is insufficient. Evidence-sufficiency research highlights the need to calibrate answer-abstention behavior rather than treating generation as mandatory [6].

G. Positioning of This Study

This perspective is directly relevant to personal memory systems, where a question can be topically related to stored material even when the requested fact is absent from the corpus. PAM's abstention and answerability experiments operationalize this distinction by evaluating not only whether retrieval is relevant, but whether the available evidence supports answering.

III. PAM SYSTEM ARCHITECTURE

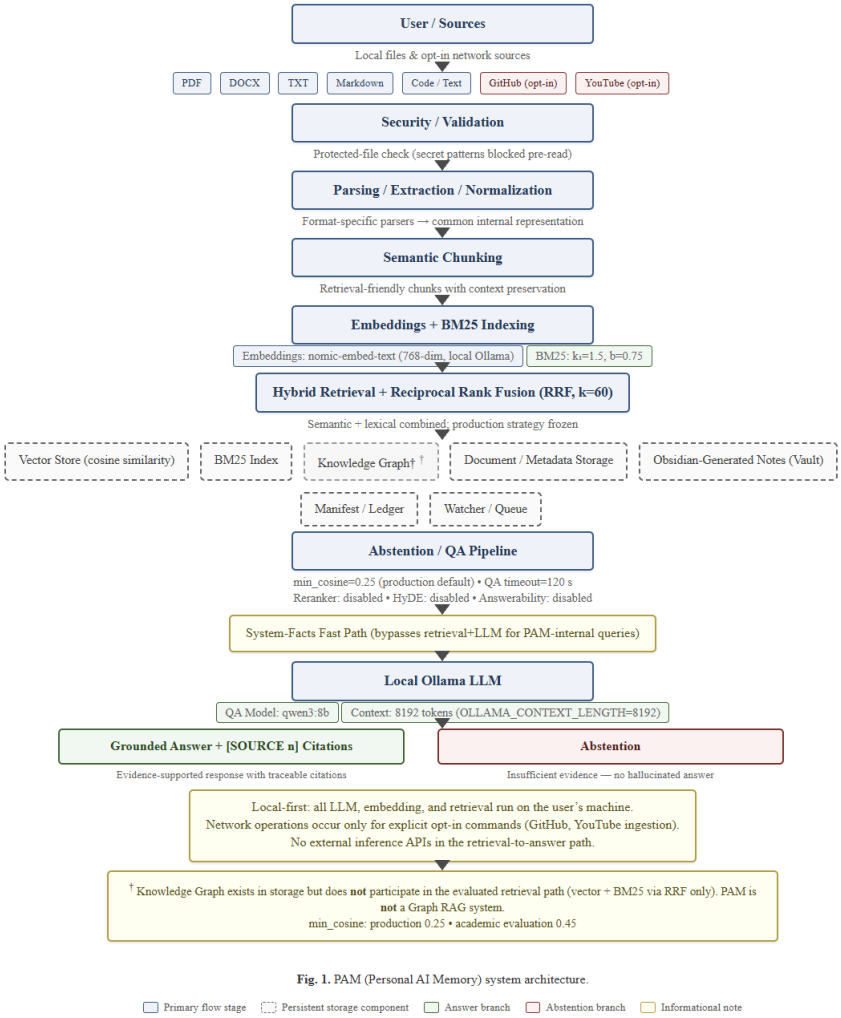


Fig. 1. PAM (Personal AI Memory) system architecture.

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```
(.venv) PS D:\Projects\Personal AI Memory> pam ask "what is Machine Learning"
[2026-09-09 12:02:23] INFO     Application started                                  logging.py:141
[2026-09-09 12:02:23] INFO     Configuration loaded                                 logging.py:142
[2026-09-09 12:02:23] DEBUG    Logging configured.                                  logging.py:143
[2026-09-09 12:02:23] INFO     Vector store loaded.                                 vector_store.py:211
[2026-09-09 12:02:26] INFO     QA retrieval completed.                              qa_workflow.py:498
[2026-09-09 12:02:26] DEBUG    Sending Ollama request.                              ollama_client.py:201
                                Answer: what is Machine Learning
Machine Learning (ML) is a subset of Artificial Intelligence (AI) that focuses on training programs to learn patterns from data rather than
relying on explicit rules. As described in the context, ML involves showing a program many examples so it can identify underlying patterns and
make predictions or decisions without being explicitly programmed for each task [SOURCE 2]. This approach contrasts with traditional
programming, where specific instructions are written for every scenario. ML is foundational to technologies like image recognition, speech
recognition, and language models, which leverage large datasets to "learn" complex relationships [SOURCE 2]. The process often involves
adjusting mathematical models (e.g., neural networks) to minimize errors through iterative training [SOURCE 5].

ANSWERED — SOURCES VERIFIED

                                Sources
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#	Document	Section	Type	Score	Snippet
2	PAM_V1_LEARNING_GUIDE.pdf	—	pdf	0.0312	PAM V1 Learning Guide A personal learning/reference guide for PAM (Personal AI Memory, package name personal-ai-memory v1.0.0). It walks from absolute beginner concepts through interview-level underst
5	But what is a neural network Deep learning chapter 1.md	Series preview	markdown	0.0285	that takes in a grid of 28x28 pixels like this and outputs a single number between 0 and 10, telling you what it thinks the digit is, well the task goes from comically trivial to dauntingly difficult.

Fig. 2. PAM ask command showing a grounded answer with [SOURCE N] citations.

A. Ingestion and Normalization

PAM accepts heterogeneous personal-knowledge sources through format-specific ingestion and normalization components. Local files are the primary source, while GitHub and YouTube are explicit network-source operations

initiated by the user. Before content processing, the ingestion path applies security and validation checks. Extracted content is normalized into retrieval-oriented representations while retaining provenance metadata such as source path, heading, and chunk index.

B. Representation and Local Storage

The system maintains several local representations of the knowledge base. Text is transformed into semantic chunks and represented by 768-dimensional embeddings generated with the local nomic-embed-text model. A BM25 index provides lexical retrieval over the same corpus. The vector store uses cosine similarity and persistent local storage rather than an external vector database in the evaluated configuration. PAM also constructs and stores a knowledge graph during ingestion; however, that graph is not queried by the evaluated retrieval path and is therefore not treated as Graph RAG in this study.

C. Hybrid Retrieval

For an ordinary knowledge query, PAM retrieves candidates using both dense similarity and BM25 lexical matching. The two ranked lists are combined using Reciprocal Rank Fusion (RRF), after which the highest-ranked candidates are passed to the downstream reliability and question-answering stages. This architecture separates retrieval from later acceptance decisions: a passage can be highly ranked because it is relevant to the query without necessarily containing the requested answer.

D. Question Answering and Abstention

The QA workflow uses the local qwen3:8b Ollama model for grounded generation. Retrieved context is assembled with source identifiers, and the prompting procedure instructs the model to ground its answer in the supplied evidence. The workflow can route PAM-internal system-facts questions through a deterministic fast path. For ordinary knowledge questions, the retrieval results are evaluated by the abstention logic before context construction and generation. Optional cross-encoder reranking, HyDE, and answerability mechanisms exist as experimental components but are disabled by default in the shipped configuration.

E. Local-First Boundary

PAM is local-first rather than universally offline. Inference in the retrieval-to-answer path is performed locally, while explicit ingestion of network sources can contact external services when the user requests those operations. This distinction is important for interpreting the system's privacy boundary. The architecture therefore emphasizes local storage and local inference without making the stronger and unsupported claim that no information can ever leave the machine.

IV. METHODOLOGY

A. Dense Retrieval

Each document chunk is represented by a 768-dimensional embedding generated locally with nomic-embed-text. Given query embedding q and chunk embedding d, dense retrieval ranks candidates by cosine similarity, sim(q,d) = (q·d)/(||q|| ||d||). The retrieval implementation uses this signal as one component of the hybrid candidate-generation stage.

$$sim(q, d) = \frac{q \cdot d}{\|q\|\|d\|}$$

B. Lexical Retrieval

Lexical retrieval is implemented using BM25 with k1=1.5 and b=0.75. BM25 ranks chunks using term frequency, inverse document frequency, and document-length normalization. Its role in the experimental design is complementary to dense retrieval: exact or overlapping terminology can remain informative even when semantic similarity is imperfect.

C. Reciprocal Rank Fusion

The dense and BM25 ranked lists are combined using Reciprocal Rank Fusion with k=60. For a document d, the fused score is RRF(d) = Σr∈R 1/(k + rankr(d)), where the summation is over the contributing ranked lists. RRF operates on rank positions rather than requiring dense and BM25 scores to be directly calibrated against one another.

$$RRF(d) = \int_r \frac{1}{k + rank_r(d)}$$

D. Candidate Selection and Context Construction

The frozen academic baseline uses top_k=5 final retrieval results. Experimental reranking stages operate on a larger candidate pool before final selection. For grounded QA, retrieved evidence is supplied with source identifiers and provenance information, and the context is bounded before generation. These settings separate candidate retrieval from the later acceptance decision.

E. Cross-Encoder Reranking

Cross-encoder reranking was evaluated as an optional second-stage intervention using a local ms-marco-MiniLM-L-12-v2 model. The cross-encoder evaluates the query and candidate passage jointly and produces a reranking score. Reranking was not assumed to improve reliability a priori; it was evaluated against the predefined guardrails. Importantly, the

reported reranker experiments used earlier evaluation corpora rather than the final frozen 199-query corpus, and are therefore reported as intervention evidence rather than as final-corpus baseline comparisons.

F. Retrieval-Based Abstention

The academic frozen configuration applies a cosine-based abstention threshold of 0.45 before downstream answer generation. A query is accepted when the retrieval evidence satisfies the configured acceptance condition; otherwise the system abstains. This mechanism is intended primarily to reject clearly out-of-scope queries while preserving positive-query recall. The threshold is an academic evaluation setting and should not be confused with the shipped runtime default, which uses a lower min_cosine value.

G. Answerability Verification

Two evidence-verification approaches were evaluated. A strict verifier was evaluated on the v2 160-query corpus. A banded verifier was evaluated on the frozen 199-query corpus and invoked additional evidence checking for an intermediate confidence band while auto-accepting higher-confidence cases. These mechanisms were assessed on false-positive reduction, false-negative rate, and latency rather than on FPR alone.

H. Evaluation Metrics

Retrieval effectiveness is measured using Hit@K, Recall@K, and Mean Reciprocal Rank (MRR). Reliability is measured using false-positive rate (FPR) = accepted negative queries / all negative queries and false-negative rate (FNR) = rejected positive queries / all positive queries. Negative rejection rate is the fraction of negative queries rejected. Latency is reported using p50, p95, and maximum observed query latency where available. FPR is not treated as an accuracy measure.

I. Evaluation Guardrails

The study evaluates interventions against explicit constraints established before the final comparison: FNR ≤ 0.033, Hit@5 ≥ 0.93, MRR ≥ 0.88, and retrieval-side p95 latency below 500 ms. For the later reliability target, FPR ≤ 0.50 was used as the desired rejection objective. An intervention is not considered successful merely because it improves one metric; it must be interpreted against the full set of relevant constraints and its evaluation dataset.

J. Experimental Comparison and Dataset Discipline

The evaluation is intentionally staged. Early retrieval and reranking experiments use the 50-query v1 corpus; later answerability experiments use a 160-query v2 corpus; the final frozen academic baseline and banded-verifier experiment use the v3.0 frozen corpus containing 199 queries, with 157 positive and 42 negative queries, over 24 sources and 195 chunks. Results from different corpora are not presented as directly interchangeable measurements. The final 199-query baseline is reproduced from the frozen evaluation artifact, while later runtime corpus drift is treated separately from the academic result.

V. EXPERIMENTAL SETUP

Metric	Value
Queries	199
Corpus	24 / 195
Hit@1	0.8408
Hit@3	0.9108
Hit@5	0.9236
Hit@10	0.9236
MRR	0.8773
FPR	0.8571
FNR	0.0000
Negative rejection	0.1429
Abstention rate	0.0302
Latency p50	37.4 ms
Latency p95	47.1 ms
Latency maximum	7299.3 ms

A. Frozen Corpus and Query Set

The final academic evaluation uses dataset v3.0, containing 199 queries: 157 positive queries for which supporting information is expected in the corpus and 42 negative queries for which the requested information should not be accepted as answerable. The corpus contains 24 sources and 195 chunks. The query set spans factoid, negative, comparison, cross-document, tricky, precise-detail, and multi-chunk categories, with easy, medium, and hard difficulty levels. Three queries were flagged in the audit as having unreliable ground truth and are retained with that caveat.

B. Frozen Baseline Configuration

The frozen academic baseline uses 768-dimensional nomic-embed-text embeddings, BM25 with k1=1.5 and b=0.75, and RRF with k=60. The academic abstention threshold is min_cosine=0.45 and the final retrieval depth is top_k=5. Cross-encoder reranking, HyDE, and answerability verification are disabled for the frozen baseline. These values define the evaluation configuration and are not a description of every shipped runtime default.

C. Evaluation Protocol

Queries are evaluated individually against the labeled corpus. Retrieval effectiveness is computed from ranked results, while reliability metrics classify acceptance or rejection under the configured abstention behavior. The frozen baseline was independently reproduced in a later experiment, providing a direct reproducibility check. Intervention results retain their original dataset version and are not merged into the frozen baseline statistics.

D. Experimental Guardrails

The principal guardrails are FNR ≤ 0.033, Hit@5 ≥ 0.93, MRR ≥ 0.88, and retrieval-side p95 latency below 500 ms. The later reliability objective targets FPR ≤ 0.50. The frozen baseline satisfies the FNR and retrieval-latency constraints, but falls marginally below the predefined Hit@5 and MRR effectiveness targets and remains substantially above the later FPR objective. These constraints make false-negative control a first-class requirement: reducing false positives by rejecting a substantial fraction of valid queries is not considered an acceptable reliability improvement.

E. Reproducibility and Corpus Drift

The frozen baseline is reproducible when the frozen corpus and evaluation configuration are restored. The audited frozen state contains 24 sources and 195 chunks, whereas the subsequent live vector store contained 30 sources and 299 entries. Current runtime status is therefore not used as evidence for the academic baseline. This separation prevents later ingestion activity from changing the corpus denominator or retrieval population of the reported frozen result.

F. Runtime Environment and Measurement Scope

PAM performs model inference locally through Ollama in the evaluated retrieval-to-answer path. Retrieval latency and downstream answer-generation latency are treated as distinct measurements; the principal performance guardrail concerns retrieval-side p95 latency. Reported latency statistics are taken from the corresponding evaluation runs, with the maximum value shown separately rather than treated as representative normal retrieval latency.

VI. RESULTS

Metric	Frozen v3.0 baseline
Queries	199 (157 positive / 42 negative)
Corpus	24 sources / 195 chunks
Hit@1	0.8408
Hit@3	0.9108
Hit@5	0.9236
Hit@10	0.9236
MRR	0.8773
FPR	0.8571 (36/42)
FNR	0.0000
Negative rejection	0.1429 (6/42)
Abstention rate	0.0302
Latency p50	37.4 ms
Latency p95	47.1 ms
Latency maximum	7299.3 ms

TABLE I. FROZEN V3.0 BASELINE RESULTS

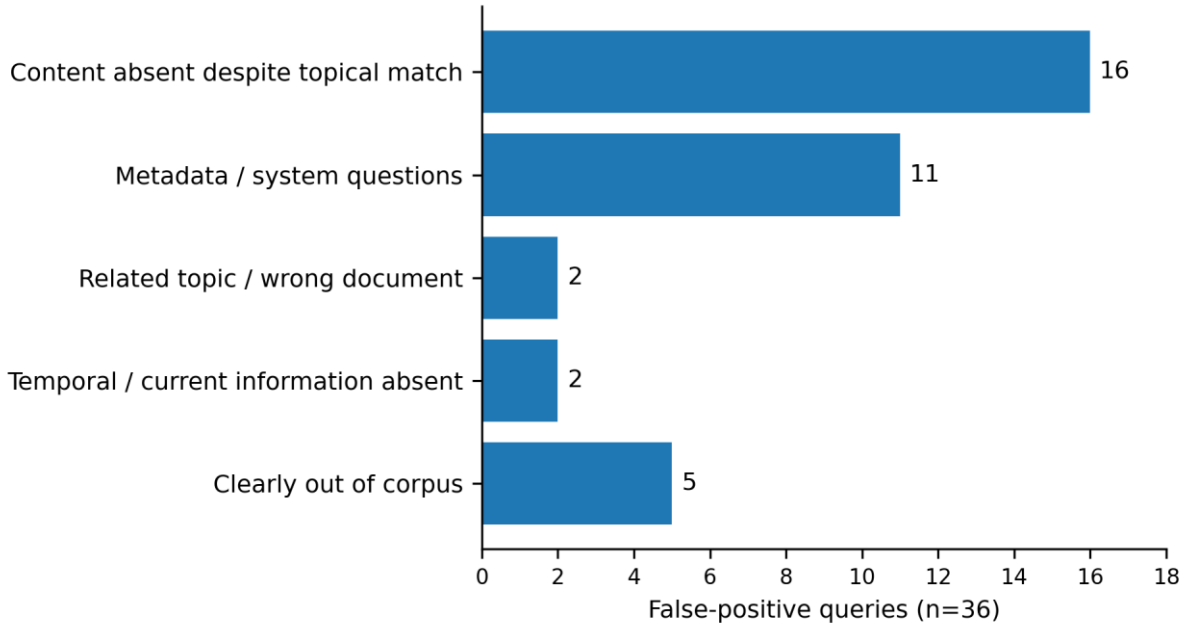


Fig. 3. Failure analysis of false positives among negative queries.

TABLE II

Mechanism	Dataset	FPR	FNR	Latency	Decision
Frozen hybrid + cosine gate	199q	0.8571	0.000	47.1 ms p95	Baseline
Cross-encoder, no gate	50q v1	1.000	0.000	~850 ms	Rejected
Cross-encoder-only gate,	160q v2	0.351	0.203	~644 ms	Rejected
Strict evidence verifier	160q v2	~0.243	~0.098	~2.9 s avg	Rejected
Banded evidence verifier	199q frozen	0.405	0.070	14.1 s avg; ~17.3 s p95	Rejected

TABLE II. SUMMARY OF MEASURED RELIABILITY INTERVENTIONS. The datasets differ across experiments; values are not treated as a single head-to-head final-corpus comparison.

A. Frozen Hybrid Retrieval Baseline

The frozen v3.0 baseline demonstrates strong retrieval performance on positive queries while providing limited rejection of negative queries. Hit@1 is 0.8408, Hit@3 is 0.9108, and Hit@5 is 0.9236; Hit@10 is also 0.9236. MRR is 0.8773. The positive-query FNR is 0, indicating that none of the 157 positive queries was rejected by the evaluated abstention configuration. However, the FPR is 0.8571 (36 of 42 negative queries), and only 6 of 42 negative queries are rejected. Retrieval latency is low, with p50=37.4 ms and p95=47.1 ms.

B. Reproducibility of the Baseline

The frozen baseline was reproduced in a later evaluation using the same frozen corpus and configuration, matching the authoritative baseline values. This supports the claim that the reported retrieval measurements are reproducible under the specified evaluation state. The reproduction is distinct from the later live runtime corpus, which had grown beyond the frozen 24-source, 195-chunk state.

C. Failure Analysis of Negative Queries

The 36 false positives were manually classified to determine why high-ranking evidence was accepted for negative queries. Five were clearly out-of-corpus requests. Sixteen concerned information that was absent even though retrieved material was topically related. Eleven were metadata or system-oriented questions, two involved a related topic or wrong document, and two required temporal or current information absent from the corpus. Thus, 31 of 36 false positives fell within the study's content-sufficiency failure analysis; this classification includes the 11 metadata/system-oriented cases, which may also be interpreted as scope failures rather than purely content-sufficiency failures.

D. Cross-Encoder Reranking

Cross-encoder reranking did not establish a reliable improvement under the evaluated constraints. On the earlier 50-query v1 evaluation, Hit@1 decreased from 0.907 to 0.837 and the corrected positive-query MRR decreased from 0.930 to 0.878, while Hit@5 remained 0.953. The reranker therefore did not improve the principal retrieval metrics in that experiment. The earlier contemporaneous MRR figures (0.940 to 0.895) included negative-query effects and are not directly comparable to the frozen positive-query MRR. Threshold sweeps on the later v2 corpus demonstrated an FPR/FNR trade-off: at a threshold of 0.125, FPR was approximately 0.351 but FNR rose to approximately 0.203. Because the reranker experiments were not performed on the final frozen 199-query corpus, these results are reported as intervention evidence rather than as a replacement for the frozen baseline.

E. Evidence-Verification Interventions

The strict answerability verifier reduced false-positive acceptance on the v2 corpus to approximately 0.243, but FNR increased to approximately 0.098 and average latency was about 2.9–3.0 seconds. The banded verifier, evaluated on the frozen 199-query corpus, reduced FPR to 0.405 but produced FNR=0.070 and approximately 14.1 seconds average latency, with p95 latency around 17.3 seconds. These results show that stronger evidence checking can reject more unsupported queries, but the observed configurations violate the study's false-negative and latency constraints.

F. Overall Reliability Result

Taken together, the experiments show a consistent boundary between retrieval quality and answerability. The frozen hybrid retriever is fast and retrieves relevant information reliably, but the retrieval signals do not provide sufficient evidence that the requested fact is present. Reranking and evidence verification can change the acceptance trade-off, but the evaluated interventions do not satisfy the complete guardrail bundle. The results therefore support the study's central finding that the dominant reliability problem lies in determining evidence sufficiency rather than merely improving topical retrieval.

VII. DISCUSSION

A. Retrieval Relevance Is Not Answer Sufficiency

The frozen baseline demonstrates that strong retrieval ranking and reliable positive-query recall do not imply reliable negative-query rejection. Hit@5 reaches 0.9236 and FNR remains 0, yet 36 of 42 negative queries are accepted. The failure taxonomy explains this apparent contradiction: many negative queries retrieve passages that are semantically or lexically related to the requested topic, but those passages do not contain the specific fact required by the query. Retrieval therefore answers the question "what is related?" more reliably than the question "is the requested answer present?"

B. Why Cosine Abstention Has a Limited Boundary

The cosine threshold is useful for clearly out-of-scope queries because unrelated queries can receive weaker retrieval evidence. However, topical adjacency produces high enough similarity to pass the same threshold even when the requested fact is absent. The 31/36 content-sufficiency false positives demonstrate that this is not primarily a threshold-calibration problem: the system can be confidently relevant to the topic while remaining unable to support the requested answer. Lowering or raising a retrieval threshold alone therefore cannot fully resolve the boundary.

C. Reranking Improves a Different Objective

Cross-encoder reranking evaluates query-passage relevance more directly than the first-stage hybrid ranker, but the measured results show that improved second-stage relevance scoring does not automatically translate into improved reliability. On the early v1 corpus, reranking reduced Hit@1 and MRR. On the later v2 threshold sweep, stronger rejection reduced FPR but introduced unacceptable FNR at useful thresholds. These findings suggest that reranking can refine ordering while still operating on the same underlying evidence; it does not create missing information.

D. Evidence Verification Trades False Positives for False Negatives and Latency

The evidence-verification experiments more directly target answer sufficiency and consequently reduce false-positive acceptance more than retrieval-only interventions. However, both evaluated approaches incur substantial costs. The strict verifier lowers FPR on v2 but exceeds the allowed FNR and introduces multi-second latency. The banded verifier lowers frozen-corpus FPR to 0.405, but FNR rises to 0.070 and average latency reaches approximately 14.1 seconds. Under the study's guardrails, the problem is therefore not simply finding a verifier that rejects more queries; it is finding one that can do so without rejecting valid answers or making the system impractically slow.

E. Implications for Local-First Personal RAG

For personal-memory systems, the findings imply that reliability evaluation should distinguish at least three cases: clearly irrelevant queries, relevant queries with answer-bearing evidence, and relevant queries for which the requested information is absent. A retrieval score is naturally suited to the first distinction but is insufficient for the second and third. This suggests that future reliability mechanisms should evaluate evidence sufficiency explicitly while preserving the low-latency retrieval layer rather than repeatedly strengthening retrieval ranking alone.

F. Answers to the Research Questions

RQ1: Hybrid dense-plus-lexical retrieval provides strong positive-query retrieval performance, with frozen Hit@5=0.9236 and MRR=0.8773. RQ2: retrieval-based abstention rejects some clearly irrelevant queries without false-negative loss in the frozen evaluation, but it does not reliably reject queries whose requested information is absent from a related topic. RQ3: reranking and evidence verification improve selected reliability measures but, in the evaluated configurations, fail the combined FNR, effectiveness, and latency constraints. RQ4: the dominant failure pattern identifies a boundary between retrieval relevance and answer sufficiency, with content-sufficiency failures accounting for 31 of 36 false positives.

VIII. LIMITATIONS AND THREATS TO VALIDITY

A. Corpus Scope and Generalization

The evaluation is based on a personal-memory corpus rather than a large public benchmark. The frozen corpus contains 24 sources and 195 chunks, and the query set contains 199 labeled queries. Consequently, the measured retrieval and reliability values should be interpreted as evidence for this controlled evaluation rather than as estimates of general RAG performance across domains, users, or corpora.

B. Dataset and Ground-Truth Limitations

The query set is manually constructed and includes positive and negative cases intended to probe retrieval and answerability behavior. Three queries were flagged during the audit as having unreliable ground truth. This introduces uncertainty into category-level interpretation and limits how strongly individual query outcomes should be generalized. The study retains these cases rather than silently altering the frozen dataset.

C. Cross-Experiment Comparability

Not all interventions were evaluated on the final frozen 199-query corpus. Reranking results include experiments on the 50-query v1 corpus and threshold-sweep results on the 160-query v2 corpus. The strict verifier was also evaluated on v2, while the banded verifier was evaluated on the frozen 199-query corpus. Because these datasets differ, intervention metrics are interpreted within their respective experiments rather than treated as a single controlled leaderboard.

D. Corpus Drift

The live PAM runtime changed after the academic corpus was frozen. The audited live vector store contained 30 sources and 299 entries, compared with 24 sources and 195 chunks in the frozen evaluation state. This drift is a practical limitation of evaluating a mutable personal-memory system. The paper addresses it by treating the frozen evaluation artifact as authoritative and reporting current runtime state separately.

E. System and Architecture Scope

The knowledge graph is stored by PAM but does not participate in the evaluated retrieval path. The conclusions therefore do not establish the effectiveness of graph-based retrieval. Similarly, the study evaluates a local-first implementation and specific local models and configurations; different embedding models, language models, rerankers, or retrieval stores may produce different results.

F. Latency and Hardware Dependence

Latency measurements depend on the local execution environment, model loading state, and configuration. The reported retrieval p95 of 47.1 ms demonstrates the measured behavior of the frozen setup rather than a hardware-independent guarantee. The multi-second costs observed for answerability verification are likewise implementation- and environment-dependent, although their order of magnitude is relevant to the studied reliability trade-off.

G. Intervention Coverage

The study evaluates several retrieval and answering interventions, including abstention, cross-encoder reranking, and evidence verification, but does not exhaust the entire design space of answerability methods. The conclusion that the tested retrieval-side levers were insufficient applies to the measured experiments and guardrails, not to every possible future retrieval or verification technique.

IX. CONCLUSION

This study evaluated reliability in a local-first personal-memory RAG system by separating retrieval relevance from answer sufficiency. The frozen hybrid dense-plus-lexical baseline achieved Hit@5=0.9236, MRR=0.8773, FNR=0, and retrieval p95 latency of 47.1 ms, demonstrating that the retrieval layer can provide strong positive-query performance with low latency. At the same time, FPR remained 0.8571, showing that relevance-based retrieval and cosine abstention do not reliably determine whether the requested information is actually present in the knowledge base.

Failure analysis identified content sufficiency as the dominant category in the study's false-positive taxonomy: 31 of 36 false positives fell within the content-sufficiency analysis, including cases where evidence was topically related but insufficient to support the requested answer. The taxonomy also includes metadata/system-oriented cases that can reasonably be viewed as scope failures. Cross-encoder reranking and evidence-verification mechanisms changed the false-positive/false-negative trade-off, but the evaluated configurations did not satisfy the complete reliability and latency guardrails.

The resulting contribution is an empirical characterization of a reliability boundary in personal-memory RAG. The findings suggest that improving retrieval ranking alone is insufficient when the central failure is absence of answer-bearing evidence. Future work should therefore investigate evidence-sufficiency mechanisms that preserve the low latency and low false-negative behavior of the retrieval layer while improving rejection of unsupported queries.

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APPENDIX A

PAM CLI AND OPERATIONAL SCREENSHOTS

This appendix provides visual evidence of the implemented PAM command-line interface and operational workflow. The screenshots are demonstrations of the current runtime and are not used to compute the frozen academic evaluation metrics.

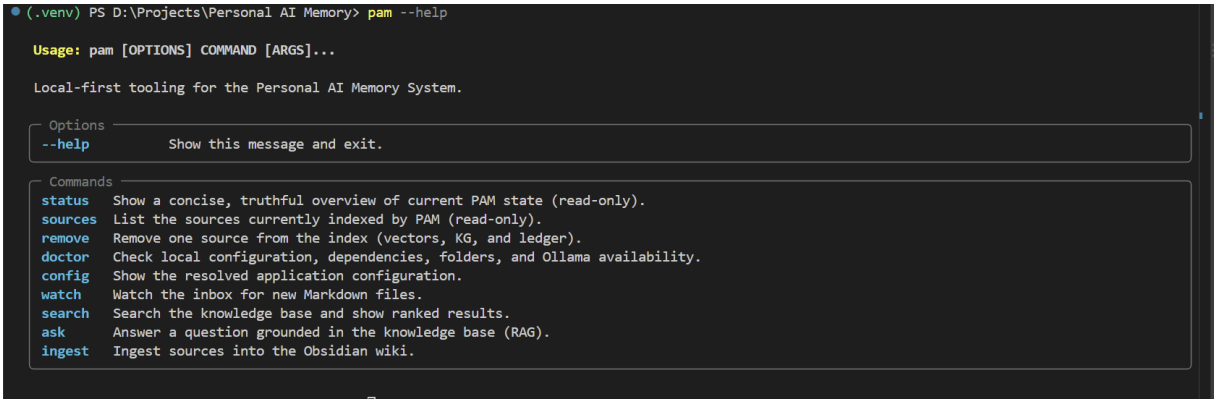


Fig. A1. PAM command-line help showing the available operational commands.

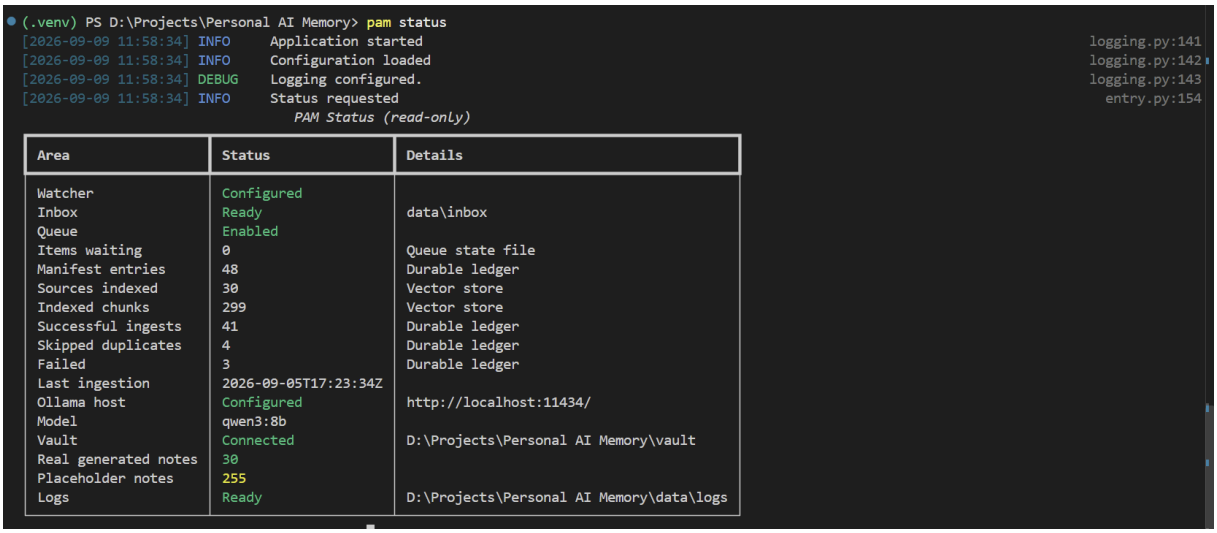
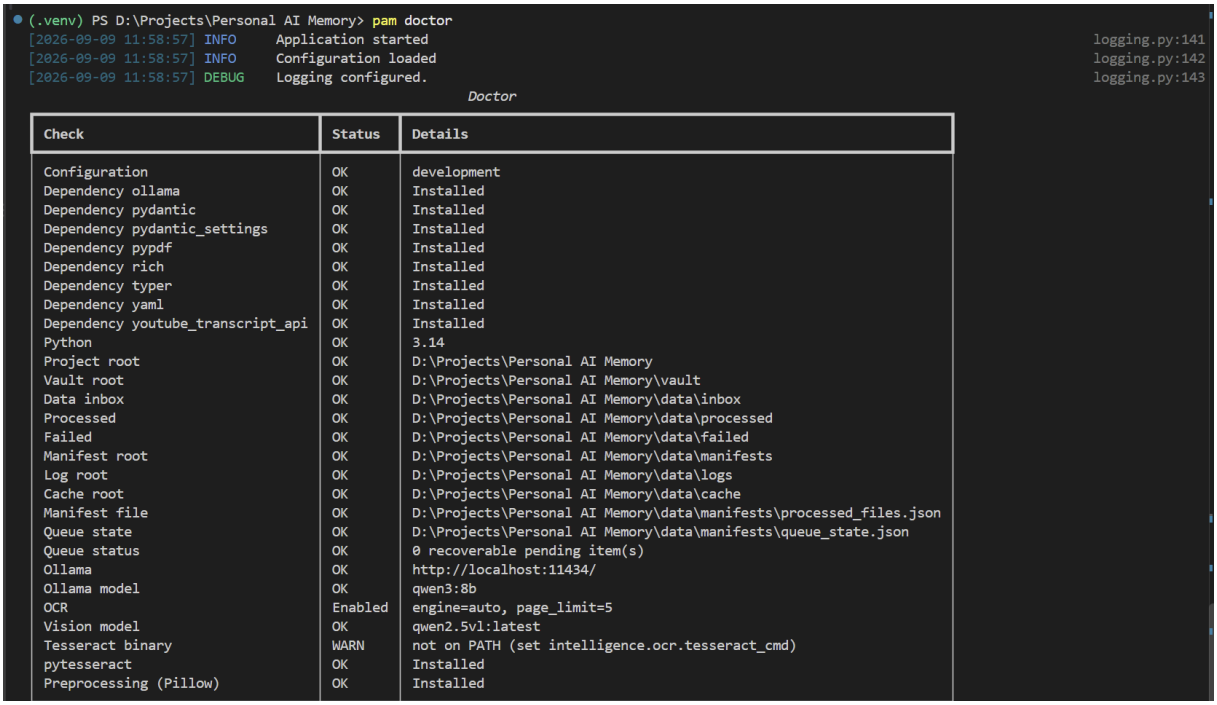


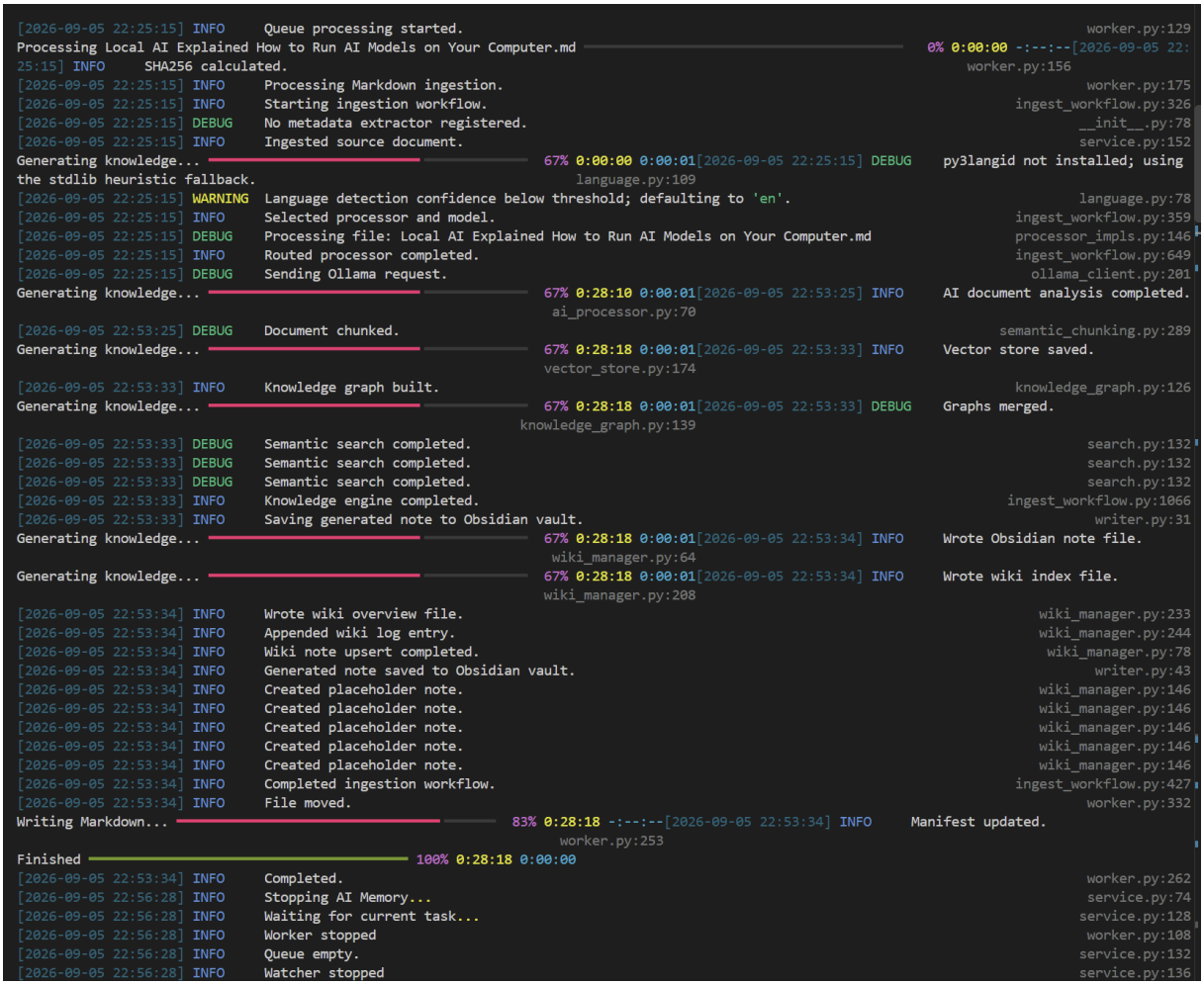
Fig. A2. PAM status output showing the current local runtime state.



```
(.venv) PS D:\Projects\Personal AI Memory> pam doctor
[2026-09-09 11:58:57] INFO     Application started                                                    logging.py:141
[2026-09-09 11:58:57] INFO     Configuration loaded                                                   logging.py:142
[2026-09-09 11:58:57] DEBUG    Logging configured.                                                    logging.py:143
                                                Doctor
```

Check	Status	Details
Configuration	OK	development
Dependency ollama	OK	Installed
Dependency pydantic	OK	Installed
Dependency pydantic_settings	OK	Installed
Dependency pypdf	OK	Installed
Dependency rich	OK	Installed
Dependency typer	OK	Installed
Dependency yaml	OK	Installed
Dependency youtube_transcript_api	OK	Installed
Python	OK	3.14
Project root	OK	D:\Projects\Personal AI Memory
Vault root	OK	D:\Projects\Personal AI Memory\vault
Data inbox	OK	D:\Projects\Personal AI Memory\data\inbox
Processed	OK	D:\Projects\Personal AI Memory\data\processed
Failed	OK	D:\Projects\Personal AI Memory\data\failed
Manifest root	OK	D:\Projects\Personal AI Memory\data\manifests
Log root	OK	D:\Projects\Personal AI Memory\data\logs
Cache root	OK	D:\Projects\Personal AI Memory\data\cache
Manifest file	OK	D:\Projects\Personal AI Memory\data\manifests\processed_files.json
Queue state	OK	D:\Projects\Personal AI Memory\data\manifests\queue_state.json
Queue status	OK	0 recoverable pending item(s)
Ollama	OK	http://localhost:11434/
Ollama model	OK	qwen3:8b
OCR	Enabled	engine=auto, page_limit=5
Vision model	OK	qwen2.5vl:latest
Tesseract binary	WARN	not on PATH (set intelligence.ocr.tesseract_cmd)
pytesseract	OK	Installed
Preprocessing (Pillow)	OK	Installed

Fig. A3. PAM doctor output validating local configuration, dependencies, and runtime readiness.



```
[2026-09-05 22:25:15] INFO     Queue processing started.                                              worker.py:129
Processing Local AI Explained How to Run AI Models on Your Computer.md ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0% 0:00:00 -:--:--[2026-09-05 22:
25:15] INFO     SHA256 calculated.                                                                     worker.py:156
[2026-09-05 22:25:15] INFO     Processing Markdown ingestion.                                         worker.py:175
[2026-09-05 22:25:15] INFO     Starting ingestion workflow.                                  ingest_workflow.py:326
[2026-09-05 22:25:15] DEBUG    No metadata extractor registered.                                      __init__.py:78
[2026-09-05 22:25:15] INFO     Ingested source document.                                              service.py:152
Generating knowledge... ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67% 0:00:00 0:00:01[2026-09-05 22:25:15] DEBUG    py3langid not installed; using
the stdlib heuristic fallback.                                                          language.py:109
[2026-09-05 22:25:15] WARNING  Language detection confidence below threshold; defaulting to 'en'.     language.py:78
[2026-09-05 22:25:15] INFO     Selected processor and model.                                 ingest_workflow.py:359
[2026-09-05 22:25:15] DEBUG    Processing file: Local AI Explained How to Run AI Models on Your Computer.md  processor_impls.py:146
[2026-09-05 22:25:15] INFO     Routed processor completed.                                   ingest_workflow.py:649
[2026-09-05 22:25:15] DEBUG    Sending Ollama request.                                         ollama_client.py:201
Generating knowledge... ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67% 0:28:10 0:00:01[2026-09-05 22:53:25] INFO     AI document analysis completed.
                                                                                         ai_processor.py:70
[2026-09-05 22:53:25] DEBUG    Document chunked.                                      semantic_chunking.py:289
Generating knowledge... ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67% 0:28:18 0:00:01[2026-09-05 22:53:33] INFO     Vector store saved.
                                                                                        vector_store.py:174
[2026-09-05 22:53:33] INFO     Knowledge graph built.                                 knowledge_graph.py:126
Generating knowledge... ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67% 0:28:18 0:00:01[2026-09-05 22:53:33] DEBUG    Graphs merged.
                                                                                   knowledge_graph.py:139
[2026-09-05 22:53:33] DEBUG    Semantic search completed.                                             search.py:132
[2026-09-05 22:53:33] DEBUG    Semantic search completed.                                             search.py:132
[2026-09-05 22:53:33] DEBUG    Semantic search completed.                                             search.py:132
[2026-09-05 22:53:33] INFO     Knowledge engine completed.                                ingest_workflow.py:1066
[2026-09-05 22:53:33] INFO     Saving generated note to Obsidian vault.                               writer.py:31
Generating knowledge... ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67% 0:28:18 0:00:01[2026-09-05 22:53:34] INFO     Wrote Obsidian note file.
                                                                                        wiki_manager.py:64
Generating knowledge... ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67% 0:28:18 0:00:01[2026-09-05 22:53:34] INFO     Wrote wiki index file.
                                                                                        wiki_manager.py:208
[2026-09-05 22:53:34] INFO     Wrote wiki overview file.                                      wiki_manager.py:233
[2026-09-05 22:53:34] INFO     Appended wiki log entry.                                       wiki_manager.py:244
[2026-09-05 22:53:34] INFO     Wiki note upsert completed.                                     wiki_manager.py:78
[2026-09-05 22:53:34] INFO     Generated note saved to Obsidian vault.                                writer.py:43
[2026-09-05 22:53:34] INFO     Created placeholder note.                                      wiki_manager.py:146
[2026-09-05 22:53:34] INFO     Created placeholder note.                                      wiki_manager.py:146
[2026-09-05 22:53:34] INFO     Created placeholder note.                                      wiki_manager.py:146
[2026-09-05 22:53:34] INFO     Created placeholder note.                                      wiki_manager.py:146
[2026-09-05 22:53:34] INFO     Created placeholder note.                                      wiki_manager.py:146
[2026-09-05 22:53:34] INFO     Completed ingestion workflow.                                 ingest_workflow.py:427
[2026-09-05 22:53:34] INFO     File moved.                                                            worker.py:332
Writing Markdown... ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 83% 0:28:18 -:--:--[2026-09-05 22:53:34] INFO     Manifest updated.
                                                                                           worker.py:253
Finished ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:28:18 0:00:00
[2026-09-05 22:53:34] INFO     Completed.                                                             worker.py:262
[2026-09-05 22:56:28] INFO     Stopping AI Memory...                                                  service.py:74
[2026-09-05 22:56:28] INFO     Waiting for current task...                                           service.py:128
[2026-09-05 22:56:28] INFO     Worker stopped                                                         worker.py:108
[2026-09-05 22:56:28] INFO     Queue empty.                                                          service.py:132
[2026-09-05 22:56:28] INFO     Watcher stopped                                                       service.py:136
```

Fig. A4. PAM ingestion workflow showing processing, AI analysis, chunking, and vector-store persistence.

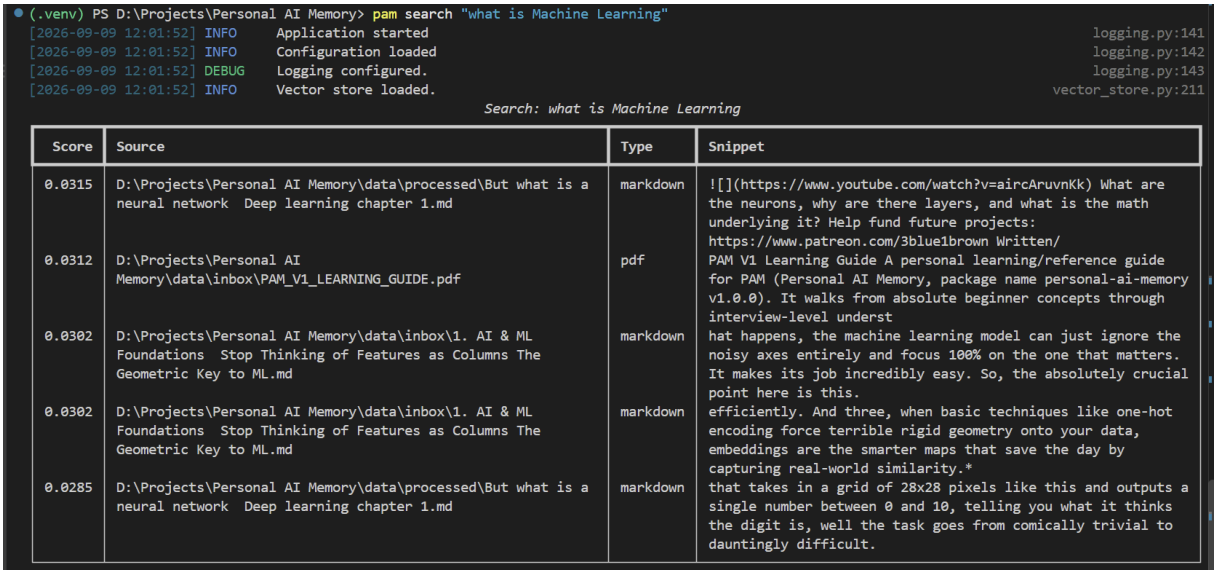


Fig. A5. PAM search command showing ranked retrieval results for a natural-language query.